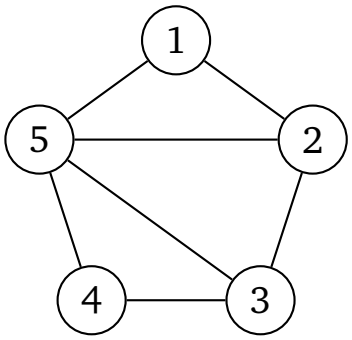
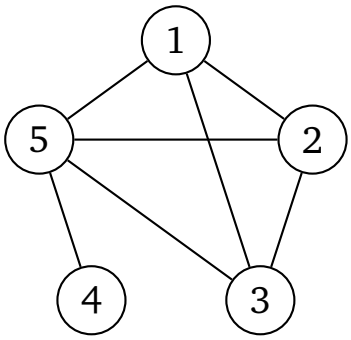
## Part 1 — How imperfect can a group be?

Here are two study groups of five. In neither group is everybody friends with everybody. Each group has seven friendships.




the Monday group




the Friday group

**Question 1**: Without counting anything, which of the two looks more like a real group to you? Say why in one sentence.

**Question 2**: Work out four numbers for each group and write them in the table.

| | the fewest friends any member has | the most members any one person is *not* friends with | friendships, out of the 10 possible | the most steps between two members |
|---|---|---|---|---|
| name | | | | |
| Monday | | | | |
| Friday | | | | |

Each column is one of the four definitions from the lecture: $k$-core, $k$-plex, $\rho$-dense and $n$-clique. Write the right name in the "name" row.

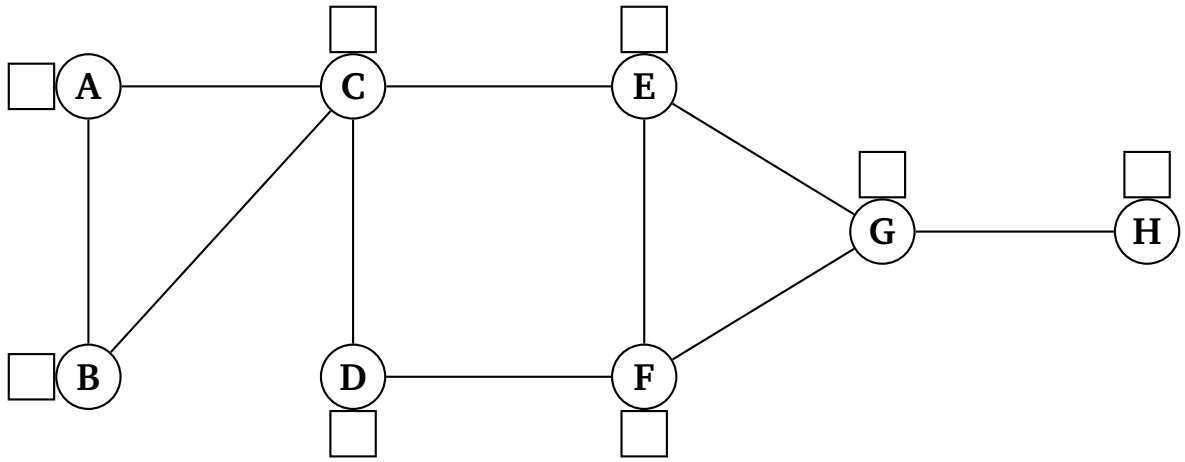
**Question 3**: Which columns say the Monday group and the Friday group are equally good groups?

Which columns tell them apart?

Look at person 4 in the Friday group. Is person 4 part of that group? Which column would you use to decide? Does your neighbour pick the same one?

## Part 2 — Two teams

You teach a lab section of eight students, A to H. The lines are the ten friendships between them. You want to split the section into two teams and keep friends together. The boxes are for Part 3; leave them empty for now.




**Question 4**: Draw one line on the drawing that splits the eight students into two teams. Cross as few friendships as you can.

How many friendships does your line cross?

**Question 5**: Compare with your neighbour. Is there a line that crosses even fewer friendships?

Would you really use that line to make two teams? Why?

**Question 6**: Here are two ways to split the section.

Split X: H alone, and the other seven together.

Split Y: A, B, C, D in one team, and E, F, G, H in the other.

For each split, count the friendships it crosses. Then work out the ratio cut from the lecture:

$$\text{ratio cut} \;=\; \frac{\text{friendships crossed}}{(\text{people in team 1}) \times (\text{people in team 2})}$$

A lower ratio cut is a better split. Which split wins now? What did dividing by the two team sizes change?

## Part 3 — More than chance

Play the game from the lecture. Every friendship is a string with a ball on each end, so the ten friendships in Part 2 make twenty balls. A student with three friends holds three balls. Paint each ball in the colour of its owner's team.

**Question 7**: In each box on the drawing in Part 2, write how many friends that student has.

**Question 8**: Score four ways to make teams. For each row:

- *observed*: friendships inside a team, divided by 10.
- *chance*: for each colour, take (balls of that colour ÷ 20), square it, and add these up over the colours.
- $Q$ = observed − chance.

| teams | friendships inside a team | balls of each colour | observed | chance | $Q$ |
|---|---|---|---|---|---|
| everybody in one team | | | | | |
| Split X: {H}, {the rest} | | | | | |
| Split Y: {A,B,C,D}, {E,F,G,H} | | | | | |
| {A,B,C,D}, {E,F,G}, {H} | | | | | |

**Question 9**: Which row has the highest $Q$?

Split X crossed the fewest friendships in Part 2. What does $Q$ say about it?

The ratio cut in Question 6 had to be told "two teams". Did anybody tell $Q$ how many teams to make?

**Question 10**: Without calculating: put every student in a team of their own, eight teams of one. What is *observed*? Is $Q$ positive or negative? Why?